Basilisk version Test

1.Introduction

The Basilisk Version Test was conducted to verify compatibility between the existing project modules and the updated Basilisk 2.8 simulation environment. This test aimed to identify any functional or structural changes that could impact the operation of previously developed scripts, particularly those integrated with BSK-RL (version 1.1.0). During the transition from Basilisk 2.2.1 to 2.8, several internal updates were introduced, including the renaming of the main Python package from “Basilisk” to “bsk.” As a result, modules such as BSK-RL, which depend on the old package naming, encounter import errors and fail to execute properly. This report documents the testing process, observed outcomes, and evidence confirming the version incompatibility between Basilisk 2.8 and older dependent modules.

2.Test Objective

The primary objective of this test was to determine whether the latest **Basilisk 2.8** release maintains compatibility with project modules originally developed under **Basilisk 2.2.1**. The focus was on identifying structural or dependency changes that affect the functionality of **BSK-RL (version 1.1.0)** and other existing scripts used for satellite fault injection and reinforcement learning simulations. Specifically, the test aimed to confirm whether BSK-RL could successfully detect and import the Basilisk package within the updated environment. By running comparative installation checks, command-line verifications, and test executions, this process sought to clearly demonstrate any incompatibility between the two versions and establish the root cause of the observed failures.

3.Test Procedure

The version compatibility test was carried out using a controlled Python environment where both Basilisk 2.2.1 and Basilisk 2.8 were independently installed and evaluated. The steps below outline the procedure followed to verify functionality and identify conflicts between the two versions:

3.1.Environment Setup:

A dedicated virtual environment named basilisk-env was activated to ensure a clean and isolated testing space. All required dependencies, including Python 3.11 and CMake 3.31.6, were preinstalled.

During setup, Conan was updated from version 1.63 to 2.3.0 to address compatibility issues with Basilisk 2.8 and ensure successful package builds.

3.2.Installation and Verification

The installation was verified using: **pip show bsk** and **pip show Basilisk**

```
C:\Users\perer\Desktop\Basilisk\basilisk\basilisk-env\Scripts>activate

(basilisk-env) C:\Users\perer\Desktop\Basilisk\basilisk\basilisk-env\Scripts>pip show bsk
Name: bsk
Version: 2.8.32
Summary: Basilisk: an Astrodynamics Simulation Framework
Home-page: https://avslab.github.io/basilisk/
Author:
Author-email:
License:
ISC License
```

```
(basilisk-env) C:\Users\perer\Desktop\Basilisk\basilisk\basilisk-env\Scripts>pip show Basilisk
WARNING: Package(s) not found: Basilisk
```

The results confirmed that only bsk (2.8.32) existed, and Basilisk was not found in the environment.

3.3.Testing BSK-RL Functionality

The existing BSK-RL 1.1.2 package was reinstalled and tested using: **pytest tests/unittest**

This process was designed to verify whether BSK-RL could successfully detect, import, and operate with the updated Basilisk package. During execution, the framework attempted to access the legacy **“Basilisk”** module that existed in previous versions, rather than the new **“bsk”** package introduced in Basilisk 2.8. As a result, multiple import errors were observed, confirming that the BSK-RL package could not interface with the newer Basilisk structure.

3.4.Observation of Errors

During testing, multiple import errors were detected where BSK-RL attempted to access the package **“Basilisk”**, which no longer exists in Basilisk 2.8. The terminal output clearly showed:

```
E   ImportError: The 'Basilisk' distribution was not found. Install from http://hanspeterschaub.info/basilisk/.
______________________________ ERROR collecting tests/unittest/sats/test_satellite.py _______________________________
ImportError while importing test module 'C:\Users\perer\Desktop\Basilisk\bsk_rl\tests\unittest\sats\test_satellite.py'.
Hint: make sure your test modules/packages have valid Python names.
Traceback:
..\..\..\AppData\Local\Programs\Python\Python311\Lib\importlib\metadata\__init__.py:563: in from_name
    return next(cls.discover(name=name))
E   StopIteration

During handling of the above exception, another exception occurred:
src\bsk_rl\check_bsk_version.py:29: in check_bsk_version
    bsk_version = parse_version(version("Basilisk"))
..\..\..\AppData\Local\Programs\Python\Python311\Lib\importlib\metadata\__init__.py:1008: in version
    return distribution(distribution_name).version
..\..\..\AppData\Local\Programs\Python\Python311\Lib\importlib\metadata\__init__.py:981: in distribution
    return Distribution.from_name(distribution_name)
..\..\..\AppData\Local\Programs\Python\Python311\Lib\importlib\metadata\__init__.py:565: in from_name
    raise PackageNotFoundError(name)
E   importlib.metadata.PackageNotFoundError: No package metadata was found for Basilisk
```

This structured procedure successfully demonstrated the package name change and the resulting incompatibility between Basilisk 2.8 and BSK-RL 1.1.2.

4.3 GUI Simulation Behavior

The graphical user interface (Vizard) launched successfully and displayed simulation visuals without major issues.

However, when attempting to inject a **battery fault**, a runtime error occurred:

Simulation Error

Error: You tried to add this variable: faultConfig
To this class: <Basilisk.simulation.spacecraft.Spacecraft; proxy of <Swig Object of type 'Spacecraft *' at 0x000001C68A66FCC0> >

OK

This issue was caused by the removal of the faultConfig variable in Basilisk 2.8.

In earlier versions (e.g., 2.2.1), this variable allowed fault configurations to attach directly to spacecraft objects, whereas the new version handles fault injections as separate modules.

5.Conclusion and Recommendations

5.1 Conclusion

The version testing confirmed that **Basilisk 2.8** introduces several changes that directly affect the compatibility of existing project modules, particularly those developed under **Basilisk 2.2.1**.

The key difference identified is the renaming of the core Python package from **"Basilisk"** to **"bsk."** This modification causes dependent modules such as **BSK-RL 1.1.2** to fail when attempting to import the old package name.
Additionally, legacy simulation variables like faultConfig, previously used to attach fault configurations directly to spacecraft objects, have been removed in the updated version. This change results in runtime errors during GUI-based operations such as battery fault injection.

Despite these compatibility issues, **Basilisk 2.8** functions correctly as an independent simulation framework, supporting modern builds, updated SPICE data, and GUI visualizations. The incompatibilities arise solely from outdated module dependencies that rely on the older Basilisk structure.

5.2 Recommendations

It is recommended to use an older and stable version of Basilisk, such as **Basilisk 2.2.1**, together with **BSK-RL version 1.1.0 or 1.1.2**, to ensure full compatibility and prevent import or runtime errors. Users should follow the provided **installation guide** to correctly set up the older Basilisk environment and use the included **requirements file** to install all necessary Python libraries before running the simulation or fault injection scenarios. This approach guarantees that the modules, GUI features, and reinforcement learning components function consistently without conflicts caused by version mismatches.